

Smart Child Presence Detection in School Buses

Problem Statement & Deliverables

1. Background

A recent alarming incident in Noida highlighted a severe safety gap in school transport systems: a UKG student was left **locked inside a school bus for nearly seven hours after falling asleep**, only to be found later crying and overheated in the parked vehicle about 25 km from the school campus. This oversight caused immense distress to the child and parents, and raised serious concerns about existing bus safety protocols and checks.

Children left unattended in buses are exposed to serious health, psychological, and life-threatening risks. Manual checks by drivers or staff have proven insufficient to consistently prevent these dangerous oversights.

News: [Times Of India](#)

2. Problem Statement

There is an urgent need for a **reliable, automated system to detect the presence of a child inside a school bus — especially after completion of a trip — and trigger timely alerts to relevant stakeholders before harm can occur.**

Current practices rely heavily on manual inspection, which can fail due to human error, fatigue, or distraction. A technological solution must ensure that no child remains unnoticed inside a vehicle after the bus completes its route, irrespective of whether bus staff remember to check.

3. Core Requirements & Considerations (Merged)

The proposed solution must address the following:

- **Automated Presence Detection:** The system should accurately detect human presence in the bus interior even when occupants are still or inactive (e.g., asleep).
- **Environmental Reliability:** It must function under varying conditions (heat inside parked bus, lighting changes, vibration, noise).
- **Low False Alerts:** Minimize incorrect alarms through robust sensing and intelligent processing.
- **Immediate Alerts:** Trigger multi-channel notifications (audible alarm in bus, SMS/app/in-system alerts to transport authorities, school administration, and caregivers).
- **Independent Operation:** Work reliably even without continuous internet — with fallback storage of logs and delayed uploads.
- **Retrofit Friendly & Cost-Effective:** Compatible with existing fleets without major modifications; easy to maintain and scalable for large bus fleets.

The system should be capable of detecting the presence of a child **before or after the bus is parked, shut down, or switched off** and must support escalation workflows if initial alerts go unacknowledged.

4. Deliverables (Submission Requirements)

All solution proposals should be submitted in **PowerPoint (PPT) format** and must include the following key components:

A. Proposed System & Architecture

- Block diagram of the solution
- Detection approach (e.g., sensor suite, algorithms)
- Data flow and alert mechanism

B. Detection Methodology

- Sensors and technologies used

- How detection accuracy is achieved
- Strategies to reduce false positives/negatives

C. Bill of Materials (BOM)

- Complete list of hardware parts
- Estimated cost per unit
- Vendor suggestions / part alternatives

D. Alerting & Notification Strategy

- Alarm triggers (in-bus, external notifications)
- Stakeholder escalation plan
- Communication methods (SMS, mobile apps, dashboards)

E. Deployment Plan

- Installation process
- Testing and validation approach
- Maintenance and calibration recommendations

F. Safety & Operational Considerations

- Power backup strategy
- Environmental tolerance (heat, dust, vibration)
- Privacy considerations (if cameras used)

5. Expected Outcome

The proposed solution must demonstrate how it will **ensure no child remains undetected in school buses**, increasing safety and reducing the risk of similar incidents in the future. The system should be practical, cost-effective, and ready for real-world adoption by schools and transport agencies.